

A Multimodal AI System for PCOS Prediction Using Ultrasound Images and Clinical Data

Abstract

Polycystic Ovary Syndrome (PCOS) is a prevalent endocrine disorder affecting women of reproductive age and is associated with infertility, metabolic complications, and long-term health risks. Accurate diagnosis remains challenging due to subjective ultrasound interpretation and heterogeneous clinical presentations. This study presents a **best-practice, decision-level multimodal artificial intelligence system** for PCOS prediction that integrates ultrasound image analysis and tabular clinical data. Due to the absence of patient-wise paired multimodal datasets, image-based and clinical-data-based models are trained independently and fused only at inference time. An EfficientNet-B0 convolutional neural network is employed for ultrasound image classification, while classical machine learning models are used for tabular clinical features. The proposed system emphasizes transparency, bias mitigation, reproducibility, and clinical applicability, aligning strictly with BioFusion Hackathon guidelines.

1. Literature Review

Recent advances in artificial intelligence have enabled automated PCOS detection using ultrasound images, clinical features, or a combination of both. Convolutional neural networks (CNNs) have demonstrated strong performance in extracting ovarian morphological patterns from ultrasound images; however, many studies rely on small, institution-specific datasets, limiting generalizability. Tabular-based machine learning approaches using hormonal and metabolic features provide interpretability but often fail to capture visual ovarian characteristics.

Several multimodal studies attempt early or feature-level fusion, assuming patient-wise paired datasets. In practice, such paired datasets are scarce, and early fusion introduces a risk of data leakage or artificial pairing. Moreover, many prior works lack transparent bias discussion, reproducibility details, or real-world deployment considerations.

This work addresses these gaps by proposing a **decision-level multimodal architecture** that respects data constraints, ensures fair evaluation, and reflects real clinical workflows. The system avoids automated pipelines and AutoML tools, prioritizing manual model design and interpretability.

2. Problem Identification

2.1 Clinical Background

PCOS affects approximately 5 -15% of women globally and is characterized by irregular menstruation, hyperandrogenism, and polycystic ovarian morphology. Delayed or inaccurate diagnosis increases the risk of infertility, type-2 diabetes, cardiovascular disease and endometrial cancer.

2.2 Problem Statement

PCOS diagnosis typically follows the Rotterdam criteria, requiring two of three diagnostic features. However, ultrasound interpretation is subjective, diagnostic accuracy varies across clinicians and specialist access is limited in rural and low-resource settings.

2.3 Proposed Solution

This project proposes a **decision-level multimodal PCOS prediction system** that:

- Analyzes ovarian ultrasound images using deep learning
- Processes clinical and biochemical data using classical machine learning
- Combines predictions at inference time through weighted fusion

Fusion Formula:

$$P_{final} = w_{image} \times P_{image} + w_{tabular} \times P_{tabular}$$

Where:

- P_{image} = Probability from image model
- $P_{tabular}$ = Probability from tabular model
- $w_{image}, w_{tabular}$ = Weights (sum to 1.0)

Methodological Note:

Due to the absence of patient-wise paired multimodal datasets, unimodal models are trained independently and integrated at decision level during inference.

3. Dataset Justification

3.1 Dataset Sources

- **Ultrasound Images:** Publicly available PCOS ultrasound dataset
- **Tabular Clinical Data:** Open-source PCOS clinical dataset in Excel/CSV format

3.2 Dataset Compliance Statement

All datasets used in this study are **open-source and publicly accessible**.

<https://www.kaggle.com/datasets/michaelmendiolas/pcos-clinical-dataset>

https://www.kaggle.com/datasets/vishnuvamsi05799/polycystic-ovary-syndrome-pcos?select=PCOS_Dataset

<https://www.kaggle.com/datasets/shnotweta/2000-images-of-ultrasound-for-pcos>

<https://www.kaggle.com/datasets/anaghachoudhari/pcos-detection-using-ultrasound-images>

3.3 Dataset Characteristics

- **Image Data:** PCOS and Non-PCOS ultrasound images, resized to 224×224 pixels
- **Tabular Data:** Demographic, hormonal, metabolic, and symptomatic features

- **Target Variable:** Binary PCOS diagnosis (PCOS / Non-PCOS)

4. Methodology

4.1 Data Preprocessing Pipeline

Image Preprocessing:

- Gaussian blur for speckle noise reduction , CLAHE for contrast enhancement
- Edge sharpening to highlight follicular boundaries, Normalization to [0,1]
- Data augmentation applied only to training data

Tabular Preprocessing:

- Removal of identifiers and irrelevant columns
- Median imputation for numerical features
- Mode imputation for categorical features
- One-hot encoding for categorical variables
- StandardScaler normalization fitted only on training data

Class imbalance is handled using class weights and stratified sampling.

4.2 Model Architecture

Image Model:

EfficientNet-B0 pretrained on ImageNet, adapted with:

- Global Average Pooling ,Batch Normalization ,Multiple Dropout layers
- Dense layers with L2 regularization ,Sigmoid output for binary classification

Tabular Models:

- Logistic Regression (baseline) ,Gradient Boosting ,Random Forest (constrained depth)

4.3 Training Process

- Two-phase training for image model:
 - Phase 1: Frozen backbone (feature extraction)
 - Phase 2: Fine-tuning last 10 layers with low learning rate
- Tabular models trained using stratified 5-fold cross-validation
- Early stopping and learning-rate scheduling applied

4.4 Validation Strategy

- Separate train, validation, and test sets

- Stratified sampling preserves class distribution
- Test sets remain untouched until final evaluation
- Strict data leakage prevention measures applied

5. Pretrained Model Usage & Adaptation

5.1 Rationale

Pretrained models improve convergence and generalization when medical datasets are limited. EfficientNet-B0 was selected for its balanced architecture and parameter efficiency.

5.2 Architectural Modifications

The original ImageNet classification head is replaced with custom dense layers and a binary sigmoid output. Additional regularization layers are introduced to mitigate overfitting.

5.3 Training Strategy

- Feature extraction followed by conservative fine-tuning
- Very low learning rate during fine-tuning to preserve pretrained representations

5.4 Risk & Bias Discussion

EfficientNet-B0 is pretrained on natural images, creating a domain mismatch with ultrasound imaging. This risk is mitigated through contrast enhancement, augmentation, conservative fine-tuning, and explicit limitation reporting. Potential biases related to dataset source, equipment variability, and population diversity are acknowledged.

6. Results

All quantitative metrics reported in this section are generated directly from the submitted notebook during execution.

Metrics include:

- Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion matrices, ROC curves

Error analysis highlights false positives caused by noise-like cyst patterns and false negatives in early-stage PCOS cases.

7. Real-World Application

The proposed system functions as a **clinical decision-support tool**, assisting clinicians rather than replacing them. It can be deployed in hospitals, fertility clinics, and rural healthcare centers to support early PCOS screening and reduce diagnostic variability.

8. Marketing & Impact Strategy

The solution targets public hospitals, fertility clinics, and primary healthcare centers. Its lightweight architecture enables deployment on low-cost hardware, improving accessibility in low-resource settings. The system offers practical benefits such as reduced clinician workload, standardized screening, and improved early diagnosis, supporting scalable healthcare adoption.

9. Limitations

Key limitations include unpaired multimodal data, limited dataset diversity, absence of external validation, and lack of explainability tools. These are acknowledged transparently and addressed in future work planning.

10. Future Improvements

Future directions include:

- Collecting patient-wise paired multimodal datasets
- Incorporating explainability techniques such as Grad-CAM and SHAP
- Multi-center validation studies
- Prospective clinical trials and regulatory approval pathways

Conclusion

InnoX provides an AI-powered, multimodal decision-support system that addresses critical gaps in PCOS diagnosis. By combining objective ultrasound analysis with clinical data, the system reduces diagnostic delays, improves access to care, and standardizes diagnostic quality across healthcare settings. With further validation and regulatory approval, InnoX has strong potential for real-world clinical impact.

Referneces

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